

IVANO-FRANKIVSK, Ukraine

WMO: 335260

Lat: 48.8892N

Lon: 24.6892E

Elev: 919

StdP: 14.21

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.9	4.7	-6.8	3.9	-2.1	0.7	5.9	5.4	24.7	38.2	22.4	37.6	1.6	300	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	87.4	68.7	84.0	67.8	80.8	66.3	71.3	82.2	69.6	80.3	67.9	77.9	7.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	105.6	76.9	66.0	99.1	74.3	64.4	93.6	72.6	35.7	82.7	34.1	80.2	32.8	78.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	17.7	15.5		-11.4	92.8	8.8	3.3	-17.8	95.2	-23.0	97.2	-27.9	99.0	-34.3	101.5
			DB												
			WB	-11.7	74.7	8.8	2.7	-18.0	76.7	-23.1	78.3	-28.1	79.8	-34.4	81.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	25.8	29.5	37.3	48.7	57.7	63.8	67.4	66.3	57.4	47.4	38.1	28.8
	DBStd	16.56	10.49	10.87	8.79	7.21	6.27	5.63	5.16	5.30	6.12	7.44	8.04	10.13
	HDD50	3009	749	574	399	106	11	0	0	0	10	139	363	658
	HDD65	6680	1214	993	860	491	236	89	35	49	238	546	806	1123
	CDD50	2074	0	0	4	66	250	414	539	504	231	59	7	0
	CDD65	269	0	0	0	0	10	53	108	88	10	0	0	0
	CDH74	3048	0	0	0	35	207	567	1070	972	186	11	0	0
	CDH80	859	0	0	0	6	28	121	332	330	42	0	0	0
Wind	WSAvg	5.7	5.6	6.3	7.2	6.7	5.9	5.6	5.4	4.7	4.9	5.2	5.6	5.5
Precipitation	PrecAvg	27.2	1.1	1.3	1.6	2.0	3.2	3.8	3.9	3.0	2.6	2.0	1.3	1.3
	PrecMax	40.0	2.0	3.2	4.1	4.0	6.7	9.3	10.3	6.2	5.2	5.0	2.6	2.7
	PrecMin	19.6	0.4	0.4	0.5	0.6	1.8	1.7	0.7	0.6	0.7	0.2	0.0	0.3
	PrecStd	5.3	0.4	0.5	0.9	0.7	1.2	1.8	2.2	1.4	1.3	1.3	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	57.4	66.3	78.8	83.0	86.9	91.7	91.6	84.6	75.4	64.2	53.9
		MCWB	44.8	46.7	50.4	59.5	63.6	68.4	70.3	69.4	65.8	60.6	53.5	46.1
	2%	DB	46.2	51.4	59.7	71.8	79.2	83.5	87.5	87.6	79.0	69.7	58.7	48.2
		MCWB	41.2	43.8	48.0	55.3	62.5	67.9	69.4	68.5	63.5	58.3	51.4	43.2
	5%	DB	42.5	46.6	55.1	67.7	75.5	80.5	83.8	83.8	74.8	64.4	53.7	43.2
		MCWB	38.3	41.4	45.5	53.7	61.6	66.5	68.6	67.7	62.2	55.7	48.4	39.7
	10%	DB	39.0	42.8	50.4	63.0	71.7	77.1	80.3	79.2	69.9	59.5	50.0	40.3
		MCWB	35.8	38.9	43.5	51.3	59.9	65.1	67.1	66.3	60.4	53.6	47.0	37.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.9	47.9	51.8	60.2	67.0	72.4	73.3	73.8	68.0	62.3	54.7	46.9
		MCDB	49.9	56.0	63.4	75.7	77.8	82.6	85.7	86.3	80.4	71.9	61.8	52.7
	2%	WB	41.6	44.5	49.0	57.1	64.7	69.6	71.4	71.0	65.4	59.5	52.1	43.1
		MCDB	45.4	50.0	57.9	68.8	75.4	79.4	82.8	81.8	76.1	67.8	57.6	47.0
	5%	WB	38.6	41.8	46.6	54.8	62.8	67.8	69.8	69.1	63.1	56.9	49.5	40.3
		MCDB	41.6	46.1	53.3	64.9	73.2	77.4	81.0	79.8	71.7	63.2	53.1	42.7
	10%	WB	36.0	39.1	44.1	52.6	60.7	66.1	68.1	67.3	61.0	54.1	46.8	37.7
		MCDB	38.4	42.5	49.6	61.4	69.8	75.0	77.9	77.0	68.7	59.0	49.7	39.6
Mean Daily Temperature Range	5% DB	MDBR	12.7	14.2	16.6	20.5	21.0	19.9	20.2	21.3	19.6	18.2	13.1	12.2
		MCDBR	15.8	20.0	26.4	30.4	29.2	26.9	27.7	30.2	29.9	27.8	21.3	18.0
		MCWBR	11.7	13.6	15.8	15.4	14.7	13.2	12.2	13.3	15.5	17.0	14.7	13.6
	5% WB	MCDBR	14.9	18.4	22.5	26.6	25.4	23.3	24.2	25.7	25.9	24.4	19.2	16.1
		MCWBR	11.5	13.1	14.2	14.3	13.6	12.1	11.5	12.5	14.2	15.6	13.6	12.6
Clear-Sky Solar Irradiance	taub	0.307	0.336	0.379	0.432	0.416	0.415	0.441	0.443	0.399	0.364	0.336	0.305	
	taud	2.291	2.246	2.210	2.131	2.244	2.273	2.213	2.206	2.286	2.360	2.380	2.356	
	Ebn at Noon	234	254	261	258	268	269	259	252	252	241	223	220	
	Edh at Noon	23	30	37	45	42	41	43	41	35	27	22	19	
All-Sky Solar Radiation	RadAvg	323	577	920	1320	1638	1739	1704	1539	1081	689	339	237	
	RadStd	44	76	118	135	164	131	143	143	166	98	40	34	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.31	N/A	N/A	+2.36	+1.08	+1.15	N/A	-547	+234	+86
(47)	Regional (21 neighbors)	+1.22	N/A	N/A	+1.85	+0.81	+0.81	-238	-365	+212	+90

Nomenclature: See separate page